

Consider an experiment for tossing an unfair coin which is biased to head by 30%.

⇒ ① make one simulated tossing and assume it's head.
calculate simulated probability for both head & tail and their errors

② make another one simulated tossing and assume it's tail
calculate simulated probability for both head & tail and their errors

Solution

$P(H) = 0.8$ theoretical prob, $0.5 + 0.3$

$P(T) = 0.2$ Theoretical prob. ~~0.5~~ 0.2

$$P(H)_{\text{simulated}} = 1 \quad \frac{\text{no. of repeating}}{\text{no. of trials}} = \frac{1}{1} = 1$$

$$P(T) = 0 \quad \frac{0}{1} = 0$$

$$\text{Error}_{(H)} = \left| \frac{\text{theor.} - \text{Sim}}{\text{theor.}} \right| * 100 = \left| \frac{0.8 - 1}{0.8} \right| * 100$$

$$= 25\%$$

$$\text{Error}_{(T)} = \left| \frac{0.2 - 0}{0.2} \right| \times 100 = 100\%$$

$$P(H)_{\text{Sim}} = 0$$

$$P(H) = 0.8$$

$$P(T) = 0.2$$

$$P(T)_{\text{Sim}} = 1$$

$$\text{Error}_{(H)} = \left| \frac{0.8 - 0}{0.8} \right| \times 100 = 100\%$$

$$\text{Error}_{(T)} = \left| \frac{0.2 - 1}{0.2} \right| \times 100 = 400\%$$

Question 2

A person sells insurance policy on a part-time basis. His records on the number of policies sold per week over a 50-week period are

Number Sold	Frequency	Probability	allocation
0	8	$\frac{\text{Frequency}}{\text{Total}} = \frac{8}{50}$ 0.16	00 → 15
1	15	0.30	16 → 45
2	17	0.34	46 → 79
3	7	0.14	80 → 93
4	3	0.06	94 → 99
	50		

Simulate three five-day periods, for each simulation determine the percentage of days on which two or more policies are sold

Use the following Random numbers

73, 41, 52, 63, 89

64, 20, 81, 94, 25

33, 09, 48, 08, 16

R.O

2, 1, 2, 2, 3

2, 1, 3, 4, 1

1, 0, 2, 0, 1

≥ 2

$$4/5 * 100 = 80\%$$

$$3/5 * 100 = 60\%$$

$$1/5 * 100 = 20\%$$

Exact

At least

At most

Question 3

Use one digit RN (6, 3, 5, 0, 8) to
generate Random observations for:

the color of a traffic found by a randomly
arriving car where it is green 40% of the time,
yellow 10% of the time and red 50% of the
time

Solution

green	40%.	0.4	4 → 3
yellow	10%.	0.1	4
Red	50%	0.5	5 → 9

No. of trials = 5

(6, 3, 5, 0, 8) $\xrightarrow{R.O}$ R, G, R, G, R

$$P_{\text{Sim}}(G) = \frac{2}{5} = 0.4$$

$$P_{\text{Sim}}(\text{yellow}) = \frac{0}{5} = 0$$

$$P_{\text{Sim}}(R) = \frac{3}{5} = 0.6$$

$$\text{Error}(G) = 0\%$$

$$\text{Error}(Y) = 100\%$$

$$\text{Error}(R) = 20\%$$

$$\text{Error} = \left| \frac{\text{Theoretical prob.} - \text{Sim prob.}}{\text{Theoretical prob.}} \right| \times 100$$